How Robust is Structure-Aware Retrieval? Quantifying and Mitigating the Effect of Document-Structure Errors in Retrieval-Augmented Generation

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DECLARATION

I hereby declare that the report of the P.G. Minor Project (IT897) work entitled **"How Robust is Structure-Aware Retrieval? Quantifying and Mitigating the Effect of Document-Structure Errors in Retrieval-Augmented Generation"**, submitted to the National Institute of Technology Karnataka, Surathkal, in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Information Technology**, is a bona fide record of the work carried out by me. The material in this report has not been submitted to any university or institution for the award of any degree.

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CERTIFICATE

This is to certify that the P.G. Minor Project (IT897) work entitled **"How Robust is Structure-Aware Retrieval? Quantifying and Mitigating the Effect of Document-Structure Errors in Retrieval-Augmented Generation"**, submitted by **Kshitij Khede (252IT010)** as the record of the work carried out by him, is accepted as the P.G. Minor Project (IT897) work report submission in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Information Technology** in the Department of Information Technology, NITK Surathkal, India. Duration of this minor project (IT897) is from JUNE 01, 2026 to AUGUST 15, 2026.

Prof. ANANTHANARAYANA V. S.

(Signature of the Guide with Date and Seal)

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ABSTRACT

Retrieval-Augmented Generation (RAG) increasingly represents documents not as flat text chunks but as a hierarchy of sections, so that a retriever can attach the surrounding structural context of a passage. A growing family of systems — RAPTOR, TreeRAG, HiChunk, BookRAG and layout-aware retrievers — build such a hierarchy and then *assume it is correct*. Real documents, however, are parsed imperfectly: multi-column pages are mis-ordered and section headings are mis-detected. This project asks, for the first time, how sensitive structure-aware retrieval is to such errors, whether the errors can be detected from a per-node confidence signal, and whether they can be repaired at inference time without re-parsing.

The recovered hierarchy is treated as an instrument, and two controllable error sources are injected: reading-order noise (ρ) and heading-number dropout (δ). Structural fidelity is measured with three retrieval-relevant, model-free metrics: Parent-Assignment Accuracy (PAA), Ancestor-Path Accuracy (APA, the ancestor context a hierarchical retriever would attach), and Ancestor-Context Precision (ACP). A confidence-gated repair step reattaches low-confidence nodes to a safe structural prior.

On a corpus of eleven research-paper PDFs (five seeds per setting), reading-order noise is almost harmless (APA 96.8% $\rightarrow$ 96.4% at 50% noise) because numbering-primary attachment is inherently order-robust, whereas heading-number loss degrades structure steadily (PAA 97.7% $\rightarrow$ 92.7%, APA 96.8% $\rightarrow$ 89.2% at 50% loss). Attachment errors occur in lower-confidence nodes (correct 0.61 vs. wrong 0.55; point-biserial $r = +0.16$), and confidence-gated repair recovers part of the loss (APA 92.8% $\rightarrow$ 93.3%). The results give the first robustness characterisation of structure-aware retrieval and an inference-time mitigation, and they identify learned structural confidence as the key lever for future work.

Keywords: Retrieval-Augmented Generation; Structure-Aware Retrieval; Robustness; Document Structure; Confidence Estimation; Ancestor Context.

CONTENTS

LIST OF FIGURES **iv**

LIST OF TABLES **v**

ABBREVIATIONS **vi**

1 Introduction **1**

1.1 Background 1

1.2 The Gap: Structure is Assumed Correct 1

1.3 Motivation 1

1.4 Scope and Contributions 2

1.5 Organisation of the Report 2

2 Literature Review **3**

2.1 Flat and Adaptive Retrieval 3

2.2 Structure-Aware and Hierarchical Retrieval 3

2.3 Document Layout Analysis 4

2.4 Attribution and Faithfulness 4

2.5 Outcome of Literature Review 4

2.6 Problem Statement 4

2.7 Objectives 4

2.8 Summary 5

3 Methodology **6**

3.1 Overview 6

3.2 Instrument: Layout-Aware Structure Recovery 6

3.3 Perturbation Model 6

3.4 Retrieval-Relevant Metrics 7

3.5 Confidence-Gated Repair 7

3.6 Summary 7

4 Work Done **8**

4.1 Implementation and Experimental Setup 8

4.2 Degradation Under Structural Error 8

4.3 Are Errors Predictable from Confidence? 8

4.4 Confidence-Gated Repair 9

4.5 Summary 9

5 Conclusions and Future Work **10**

5.1 Conclusions 10

5.2 Limitations 10

5.3 Future Work 10

REFERENCES **12**

LIST OF FIGURES

3.1 Study pipeline. The recovered hierarchy is an instrument; the shaded blocks (perturbation and confidence-gated repair) are the research contributions. 6

4.1 (a) Structural fidelity degrades steadily with heading-number loss δ. (b) The same hierarchy is almost unaffected by reading-order noise ρ, showing that numbering-primary attachment confers reading-order robustness. 9

LIST OF TABLES

2.1 Summary of the literature survey for the most relevant related works. . . 3

4.1 Structural fidelity (%) vs. heading-number dropout δ (mean over 11 documents, 5 seeds). 8

4.2 Confidence-gated repair ablation at $\delta = 0.3$, $\tau = 0.60$ (mean over 11 documents, 5 seeds). 9

ABBREVIATIONS

Abbreviation	Full Form
RAG	Retrieval-Augmented Generation
PDF	Portable Document Format
PAA	Parent-Assignment Accuracy
APA	Ancestor-Path Accuracy
ACP	Ancestor-Context Precision
PRE	Parent Ranking Engine
KCE	Knowledge Confidence Estimation
CPG	Candidate Parent Generation
TF–IDF	Term Frequency–Inverse Document Frequency
DFS	Depth-First Search
LLM	Large Language Model
OCR	Optical Character Recognition

CHAPTER 1

INTRODUCTION

1.1 Background

Large Language Models answer fluently but can state untrue facts and cannot see knowledge created after training. Retrieval-Augmented Generation (RAG) addresses both by retrieving relevant text from an external store at query time and giving it to the model as context [1, 2]. The quality of a RAG answer therefore depends heavily on what is retrieved, which in turn depends on how the source document is represented before retrieval.

Early RAG splits a document into flat, fixed-size chunks [3, 4]. Recent systems instead represent a document as a *hierarchy* — title, sections, subsections — and retrieve along it, so that when a passage is returned its ancestor sections are attached as context [6, 7, 8]. This is now a large and active family of "structure-aware" methods.

1.2 The Gap: Structure is Assumed Correct

Structure-aware systems all share one assumption: that the recovered hierarchy is correct. In practice the hierarchy is produced by an imperfect parser. Two failures are common on real PDFs: (i) **reading-order errors**, where a multi-column page is read across columns, and (ii) **heading-detection errors**, where a section number is missed or garbled. When a section is attached to the wrong parent, a hierarchical retriever silently attaches the *wrong ancestor context* to the query — with no error flag. No prior work quantifies how damaging this is, whether it can be predicted, or whether it can be repaired.

1.3 Motivation

If structure-aware retrieval is fragile to ordinary parsing noise, then reported gains on clean documents may not transfer to real ones, and the errors are dangerous precisely because they are silent. Conversely, if the representation already carries a usable confidence signal, that signal could be used to detect and repair weak structure cheaply, at inference time. Establishing which of these holds is a prerequisite for trusting structure-aware RAG in practice.

1.4 Scope and Contributions

This work studies robustness of the *structural representation* that feeds a hierarchical retriever; it does not train or evaluate an end-to-end generator (that is left to future work). The contributions are:

1. A controlled perturbation framework that injects two realistic, independently tunable structural error sources (reading-order noise ρ, heading-number dropout δ) into a layout-aware document pipeline.
2. Three model-free, retrieval-relevant fidelity metrics — PAA, Ancestor-Path Accuracy (APA), and Ancestor-Context Precision (ACP) — that measure the context a hierarchical retriever would attach.
3. The first empirical robustness characterisation of structure-aware retrieval, including the finding that numbering-primary attachment is inherently robust to reading-order errors while being vulnerable to heading-number loss.
4. A confidence-gated repair mechanism that reattaches low-confidence nodes at inference time, with an ablation quantifying its effect.

1.5 Organisation of the Report

Chapter 2 reviews related work and states the problem and objectives. Chapter 3 describes the instrument, the perturbation model, the metrics, and the repair. Chapter 4 reports the experiments and results. Chapter 5 concludes and lists future work.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews retrieval and document-structure methods related to this project. Table 2.1 summarises the most relevant works and the gap each leaves open.

Table 2.1: Summary of the literature survey for the most relevant related works.

Work	Category	Main idea and limitation
Lewis et al. [1]	Flat RAG	Generator + dense passage retrieval; treats documents as flat passages.
Borgeaud et al. [3]	Flat RAG	Retrieval over trillions of tokens with fixed chunks; no document structure.
Sarthi et al. [6]	Tree RAG	Recursive clustering/summarisation tree; assumes clean inputs; no structural-error study.
Tao et al. [7]	Tree RAG	Tree-chunking along headings; needs correct heading markup; robustness untested.
Lu et al. [8]	Tree RAG	Hierarchical chunking via fine-tuned LLM; evaluated on clean structure.
Zhang et al. [9]	Hierarchical index	Tree + entity graph + agent for books; assumes reliable structure extraction.
Xu et al. [10]	Layout-aware	Symbolic layout graph + agent for visually-rich docs; does not study parser-error robustness.
Wallat et al. [11]	Attribution	Shows RAG citations can be correct yet unfaithful; faithfulness under noise unsolved.

2.1 Flat and Adaptive Retrieval

Lewis et al. [1] and Guu et al. [2] connect a language model to a document store; Borgeaud et al. [3] and Izacard et al. [4] scale and jointly train retrieval. Asai et al. [5] make retrieval selective. All operate over flat chunks and discard document structure.

2.2 Structure-Aware and Hierarchical Retrieval

Sarthi et al. [6] (RAPTOR) build a tree by clustering and summarising chunks; Tao et al. [7] (TreeRAG) chunk along headings; Lu et al. [8] (HiChunk) align chunk boundaries with a learned multi-level structure. Recent systems add richer indices: Zhang et

al. [9] (BookRAG) combine a hierarchical tree with an entity graph, and Xu et al. [10] (LAD-RAG) build a symbolic layout graph queried by an agent. All of these *construct* structure and evaluate on clean documents.

2.3 Document Layout Analysis

Geometric layout analysis [12, 13] and layout-aware transformers such as LayoutLMv3 [14] recover physical structure from PDFs, but target extraction/OCR rather than the robustness of downstream retrieval.

2.4 Attribution and Faithfulness

Wallat et al. [11] show that RAG citations can be correct yet not faithful, and that comprehensive faithfulness evaluation is still open — evidence that trusting what a RAG system retrieves and cites is an unsolved, active problem.

2.5 Outcome of Literature Review

Structure-aware retrieval is now mainstream, but every reviewed method assumes the recovered structure is correct. None reports how retrieval context degrades when the structure is wrong, none localises which nodes fail, and none repairs weak structure at inference time. This is the gap the present work addresses.

2.6 Problem Statement

To quantify the sensitivity of structure-aware retrieval context to document-structure errors, and to detect and repair such errors at inference time using per-node structural confidence.

2.7 Objectives

1. To define retrieval-relevant, model-free metrics of structural fidelity (PAA, APA, ACP).
2. To build a controlled harness that injects reading-order noise (ρ) and heading-number dropout (δ) into a layout-aware document pipeline.
3. To measure how structural fidelity degrades with each error source.
4. To test whether attachment errors are predictable from per-node confidence.

5. To design and evaluate a confidence-gated repair that recovers structure at inference time.

2.8 Summary

The chapter reviewed flat, adaptive, tree, hierarchical and layout-aware retrieval and layout analysis, established that structural correctness is universally assumed, and stated the problem and objectives that follow from that gap.

CHAPTER 3

METHODOLOGY

3.1 Overview

The study has four parts: an instrument that recovers document structure, a perturbation model that corrupts it in a controlled way, retrieval-relevant metrics that score the result, and a confidence-gated repair. Figure 3.1 shows the flow; the two shaded blocks are the new research components.

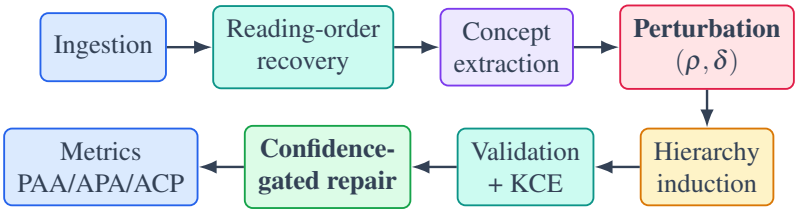


Figure 3.1: Study pipeline. The recovered hierarchy is an instrument; the shaded blocks (perturbation and confidence-gated repair) are the research contributions.

3.2 Instrument: Layout-Aware Structure Recovery

The instrument ingests a PDF with pdfplumber, recovers reading order with a column-gutter rule (blocks are split at the page gutter and sorted column by column with a vertical tolerance $\tau_y = 20$ pt), and classifies headings by section numbering or relative font size. Each heading becomes a concept node grounded to its source block. Attachment is *numbering-primary*: a number such as "3.1" attaches to "3"; unnumbered headings fall back to a Parent Ranking Engine (PRE) that scores candidates by reading-order proximity, TF–IDF similarity [15] and heading level. Every edge is validated by an upward DFS so the result is acyclic, single-parent and depth-bounded, and each node receives a Knowledge Confidence Estimation (KCE) score propagated in topological order via Kahn's algorithm [16]. This instrument is the author's prior implementation and is reused unchanged.

3.3 Perturbation Model

Two independent error sources model real parser failure. Reading-order noise swaps each adjacent pair of concepts in reading order with probability ρ. Heading-number

dropout removes a heading's section number with probability δ, forcing the semantic PRE fallback for that heading. Both are applied to copies of the extracted concepts, after which the hierarchy is rebuilt and re-scored.

3.4 Retrieval-Relevant Metrics

Let $\text{gold}(c)$ be the parent of concept c implied by its original numbering, and $\text{anc}(c)$ its gold ancestor chain to the root.

- **PAA** = fraction of nodes whose predicted immediate parent equals $\text{gold}(c)$.
- **APA** = fraction of nodes whose *entire* predicted ancestor set equals $\text{anc}(c)$; this is exactly the context a hierarchical retriever would attach.
- **ACP** = mean over nodes of $|\widehat{\text{anc}}(c) \cap \text{anc}(c)|/|\widehat{\text{anc}}(c)|$, the precision of the attached ancestor context.

No language model is used, so any change is attributable to structure alone.

3.5 Confidence-Gated Repair

Repair uses the existing KCE score as an error detector. For each non-root node c with $\text{conf}(c) < \tau$, the predicted (possibly corrupted) parent is discarded and c is reattached to the nearest earlier heading of shallower level — a safe structural prior — as in Algorithm 1. The step needs no re-parsing of the PDF.

Algorithm 1 Confidence-Gated Structural Repair

Require: concepts in reading order, confidence threshold τ

for all non-root concept c **do**
  if $\text{conf}(c) < \tau$ **then**
    $u \leftarrow$ nearest earlier concept with $\text{level}(u) < \text{level}(c)$
    $\text{parent}(c) \leftarrow u$ (or root if none)
  end if
end for

3.6 Summary

The methodology reuses a validated layout-aware instrument, corrupts its output in a controlled way, scores the result with retrieval-relevant metrics, and repairs weak structure using the existing confidence signal.

CHAPTER 4

WORK DONE

4.1 Implementation and Experimental Setup

The instrument is implemented in Python (pdfplumber, scikit-learn, NumPy, Pydantic). The new perturbation harness, metrics and repair are implemented in a single evaluation module. Experiments use a corpus of eleven real research-paper PDFs from the RAG literature (RAPTOR, GraphRAG, TreeRAG, Self-RAG, HiChunk, RETRO, Atlas, T-RAG, REALM, HAT-RAG, and the original RAG paper); gold structure is derived from section numbering. Every setting is averaged over five random seeds. No language model is involved, isolating the structural effect.

4.2 Degradation Under Structural Error

Table 4.1 and Figure 4.1 report structural fidelity as each error source increases. Reading-order noise has almost no effect (APA 96.8% $\rightarrow$ 96.4% at $\rho = 0.5$): because attachment is numbering-primary, the section number encodes the parent independently of order, so the hierarchy inherits robustness to reading-order errors. Heading-number loss degrades fidelity steadily (PAA 97.7% $\rightarrow$ 92.7%, APA 96.8% $\rightarrow$ 89.2%, ACP 97.3% $\rightarrow$ 91.2% at $\delta = 0.5$), because losing the number forces the weaker semantic fallback. This confirms H1 for heading-number loss and reveals that the dominant vulnerability is heading detection, not reading order.

Table 4.1: Structural fidelity (%) vs. heading-number dropout δ (mean over 11 documents, 5 seeds).

δ	0.0	0.1	0.2	0.3	0.4	0.5
PAA	97.7	96.3	95.1	94.7	93.8	92.7
APA	96.8	94.2	92.2	91.2	90.4	89.2
ACP	97.3	95.3	93.8	93.1	92.3	91.2

4.3 Are Errors Predictable from Confidence?

At $\delta = 0.3$, correctly attached nodes have a mean KCE confidence of 0.611 and wrongly attached nodes 0.548 (gap $+0.062$); the point-biserial correlation between confidence

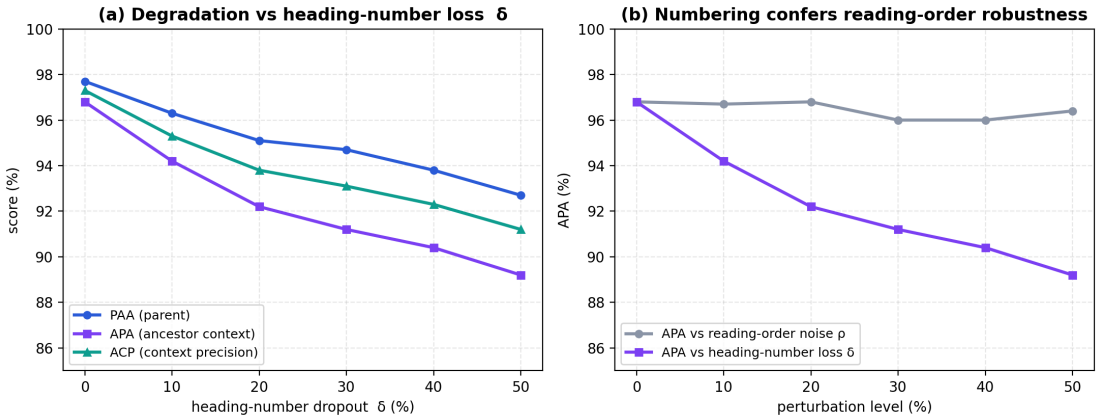


Figure 4.1: (a) Structural fidelity degrades steadily with heading-number loss δ. (b) The same hierarchy is almost unaffected by reading-order noise ρ, showing that numbering-primary attachment confers reading-order robustness.

and correctness is $r = +0.16$ over 980 nodes. Confidence is therefore a real but weak predictor of attachment error (partly supporting H2): the current heuristic KCE carries error information, but not strongly.

4.4 Confidence-Gated Repair

Table 4.2 shows the repair ablation at $\delta = 0.3$, $\tau = 0.60$. Gated repair improves ancestor-context accuracy (APA 92.8% $\rightarrow$ 93.3%) and precision (ACP 94.1% $\rightarrow$ 94.4%). The gain is positive but small, and is bounded by the weak confidence signal: the repair can only help where confidence correctly flags an error. This supports H3 while showing that the value of the mitigation is limited by confidence quality.

Table 4.2: Confidence-gated repair ablation at $\delta = 0.3$, $\tau = 0.60$ (mean over 11 documents, 5 seeds).

Setting	PAA (%)	APA (%)	ACP (%)
No repair	94.9	92.8	94.1
Gated repair	**95.1**	**93.3**	**94.4**

4.5 Summary

Across the corpus, reading-order noise is almost harmless while heading-number loss degrades the retrieval context steadily; attachment errors fall on lower-confidence nodes; and a confidence-gated repair recovers part of the loss. The size of the recovery is limited by the quality of the confidence signal.

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

5.1 Conclusions

This project gives the first robustness characterisation of structure-aware retrieval. Using a controlled perturbation of a validated layout-aware pipeline and three model-free, retrieval-relevant metrics, it shows that numbering-primary attachment is inherently robust to reading-order errors (APA $\approx 96\%$ across all ρ), while heading-number loss degrades the ancestor context that a hierarchical retriever would attach (APA $96.8\% \rightarrow 89.2\%$). Attachment errors concentrate on lower-confidence nodes, and a confidence-gated repair recovers part of the loss without re-parsing. The study turns a preprocessing pipeline into a research instrument and yields both a design insight and an inference-time mitigation.

5.2 Limitations

1. Fidelity is measured with structural proxies (PAA/APA/ACP), not end-task question answering.
2. The confidence signal is a hand-crafted heuristic and only weakly predicts errors ($r \approx 0.16$), which caps the repair gain.
3. Gold structure is numbering-derived, so unnumbered-heading correctness is not directly graded.
4. The corpus is eleven academic PDFs; non-academic layouts are untested.

5.3 Future Work

1. Replace the heuristic KCE with a *learned* structural-confidence estimator to strengthen detection and repair.
2. Add an LLM-in-the-loop evaluation to connect structural fidelity to answer quality and faithfulness.
3. Extend perturbation to the block level (before heading detection) and to larger public document-QA corpora.

4. Study robustness under heterogeneous and federated document sources.

These directions form the objectives of the Major Project, of which this robustness study is the first objective.

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